

Introduction to Fluids and Viscosity

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Fluid Mechanics Lecture Notes

Learning Objectives

By the end of this lecture, students should be able to:

- Define a fluid in the context of fluid mechanics.
- Identify common fluid properties used in fluid mechanics.
- Explain the no-slip condition.
- Derive the relationship between shear stress and velocity gradient.
- Distinguish between dynamic viscosity and kinematic viscosity.
- Solve a basic thin-fluid-film viscosity problem.

1 Introduction

Fluid mechanics is the study of fluids and the forces acting on them. In general, we can think of a fluid as either a liquid or a gas.

The definition of a fluid can change slightly depending on the source and the application. In this course, we will use the following engineering definition.

Definition

A **fluid** is a substance that continuously deforms when subjected to an external shear stress.

This definition is important because it separates fluids from solids. A solid can resist a shear stress and deform to a fixed shape. A fluid continues to deform as long as the shear stress is applied.

Examples of fluids include water, oil, air, and steam.

2 Main Principles Used in Fluid Mechanics

To solve fluid mechanics problems, we usually apply a few fundamental laws of physics:

- Conservation of mass
- Newton's second law
- First law of thermodynamics
- Second law of thermodynamics

Sometimes people say that fluid mechanics uses conservation of momentum. More precisely, we often apply **Newton's second law** to fluid motion. We also use the laws of thermodynamics when energy and heat transfer are involved.

3 Common Fluid Properties

Before analyzing fluid motion, we need several common fluid properties.

3.1 Density

Density is mass per unit volume:

$$\rho = \frac{m}{V}. \quad (1)$$

The SI unit of density is

$$\text{kg/m}^3. \quad (2)$$

3.2 Specific Volume

Specific volume is the inverse of density:

$$v = \frac{1}{\rho}. \quad (3)$$

The SI unit of specific volume is

$$\text{m}^3/\text{kg}. \quad (4)$$

3.3 Specific Weight

Specific weight is weight per unit volume:

$$\gamma = \rho g, \quad (5)$$

where g is gravitational acceleration.

The SI unit of specific weight is

$$\text{N/m}^3. \quad (6)$$

3.4 Specific Gravity

Specific gravity is the ratio between the density of a fluid and the density of water:

$$SG = \frac{\rho_{\text{fluid}}}{\rho_{\text{water}}}. \quad (7)$$

Specific gravity is dimensionless.

4 Ideal Gas Equation

For many gas-flow applications, we use the ideal gas equation:

$$P = \rho RT. \quad (8)$$

Equivalently,

$$\frac{P}{\rho} = RT, \quad (9)$$

where P is pressure, ρ is density, R is the gas constant, and T is absolute temperature.

This equation is useful when the working fluid is a gas and the ideal gas assumption is reasonable.

5 Viscosity

Viscosity is one of the most important properties in fluid mechanics. It measures a fluid's resistance to deformation or flow.

For example:

- Air has low viscosity.
- Water has higher viscosity than air.
- Oil usually has higher viscosity than water.

A more viscous fluid requires more force to shear or move.

6 Velocity Variation Near a Surface

When a fluid flows over a solid surface, the velocity of the fluid is not uniform near the surface. The fluid close to the surface moves more slowly than the fluid farther away.

For example, wind speed near the ground is lower than wind speed at higher elevations. This is one reason why wind turbines are placed high above the ground instead of directly at ground level.

The closer the fluid is to the surface, the slower it moves. Exactly at the surface, the fluid velocity is equal to the velocity of the surface.

7 No-Slip Condition

The **no-slip condition** states that a fluid in direct contact with a solid surface has the same velocity as that surface.

If the surface is stationary, then the fluid velocity at the surface is zero:

$$u = 0. \quad (10)$$

If the surface moves with velocity V , then the fluid particles touching that surface move with the same velocity:

$$u = V. \quad (11)$$

Important Rule

At a solid surface, the fluid velocity equals the velocity of the surface.

8 Fluid Between Two Parallel Plates

Consider a fluid trapped between two horizontal plates:

- The bottom plate is stationary.
- The top plate moves to the right with velocity V .
- The distance between the plates is h .

Because of the no-slip condition,

$$u = 0 \quad \text{at the bottom plate,} \quad (12)$$

and

$$u = V \quad \text{at the top plate.} \quad (13)$$

For a simple linear velocity profile, the velocity gradient is

$$\frac{du}{dy} = \frac{V}{h}. \quad (14)$$

This means that the velocity changes from zero at the stationary plate to V at the moving plate over the distance h .

9 Shear Strain Rate

Assume the top plate moves a small distance dx during a small time interval dt . Then the velocity of the top plate is

$$V = \frac{dx}{dt}. \quad (15)$$

For a small deformation angle $d\theta$,

$$\tan(d\theta) = \frac{dx}{h}. \quad (16)$$

For a very small angle,

$$\tan(d\theta) \approx d\theta. \quad (17)$$

Therefore,

$$d\theta = \frac{dx}{h}. \quad (18)$$

Dividing both sides by dt gives

$$\frac{d\theta}{dt} = \frac{1}{h} \frac{dx}{dt}. \quad (19)$$

Since $dx/dt = V$,

$$\frac{d\theta}{dt} = \frac{V}{h}. \quad (20)$$

The quantity $d\theta/dt$ is called the **shear strain rate**. For a general velocity profile, we write

$$\frac{d\theta}{dt} = \frac{du}{dy}. \quad (21)$$

The unit of shear strain rate is

$$\text{s}^{-1}. \quad (22)$$

10 Dynamic Viscosity

For a Newtonian fluid, shear stress is proportional to shear strain rate:

$$\tau = \mu \frac{du}{dy}, \quad (23)$$

where τ is shear stress, μ is dynamic viscosity, and du/dy is the velocity gradient.

This equation is called **Newton's law of viscosity**.

Newtonian Fluid

A Newtonian fluid has a linear relationship between shear stress and shear strain rate:

$$\tau = \mu \frac{du}{dy}.$$

For these fluids, μ is constant at a given temperature and pressure.

Examples of Newtonian fluids include air, water, and many simple oils. Some fluids are non-Newtonian, meaning their apparent viscosity changes with shear rate. Examples include ketchup, toothpaste, and some paints.

11 Solid Mechanics Analogy

In solid mechanics, for elastic deformation, shear stress and shear strain are related by

$$\tau = G\gamma, \quad (24)$$

where G is the shear modulus and γ is shear strain.

In fluid mechanics, for a Newtonian fluid, shear stress and shear strain rate are related by

$$\tau = \mu \frac{du}{dy}. \quad (25)$$

The analogy is that both equations describe resistance to deformation. In solids, the material property is the shear modulus G . In Newtonian fluids, the material property is the dynamic viscosity μ .

12 Shear Stress and Force

Shear stress is force divided by area:

$$\tau = \frac{F}{A}. \quad (26)$$

Combining this with Newton's law of viscosity gives

$$\frac{F}{A} = \mu \frac{du}{dy}. \quad (27)$$

For a linear velocity profile between two plates,

$$\frac{du}{dy} = \frac{V}{h}. \quad (28)$$

Therefore,

$$\frac{F}{A} = \mu \frac{V}{h}. \quad (29)$$

Solving for the force,

$$F = A\mu \frac{V}{h}. \quad (30)$$

Key Equation

For a thin fluid film between a moving surface and a stationary surface,

$$F = A\mu \frac{V}{h}.$$

13 Units of Dynamic Viscosity

Starting from Newton's law of viscosity,

$$\tau = \mu \frac{du}{dy}. \quad (31)$$

Solving for μ ,

$$\mu = \frac{\tau}{du/dy}. \quad (32)$$

The unit of shear stress is

$$\text{Pa} = \text{N/m}^2. \quad (33)$$

The unit of shear strain rate is

$$\text{s}^{-1}. \quad (34)$$

Therefore,

$$\mu = \frac{\text{N/m}^2}{\text{s}^{-1}}. \quad (35)$$

Thus,

$$\mu = \text{N} \cdot \text{s/m}^2. \quad (36)$$

Since $1 \text{ Pa} = 1 \text{ N/m}^2$, dynamic viscosity can also be written as

$$\mu = \text{Pa} \cdot \text{s}. \quad (37)$$

Another equivalent unit is

$$\mu = \text{kg}/(\text{m} \cdot \text{s}). \quad (38)$$

14 Kinematic Viscosity

Kinematic viscosity is defined as dynamic viscosity divided by density:

$$\nu = \frac{\mu}{\rho}, \quad (39)$$

where ν is kinematic viscosity, μ is dynamic viscosity, and ρ is density.

The unit of kinematic viscosity is

$$\text{m}^2/\text{s}. \quad (40)$$

Dynamic viscosity measures resistance to shearing motion. Kinematic viscosity compares this resistance to the density of the fluid.

15 Worked Example: Snowmobile on a Thin Water Film

A snowmobile is traveling over an icy surface instead of snow. Between the sleds and the ice, there is a very thin film of water.

The water film has a thickness of

$$h = 14 \mu\text{m}, \quad (41)$$

and the total contact area between the sleds and the water is

$$A = 960 \text{ cm}^2. \quad (42)$$

The snowmobile is moving at a constant speed of

$$V = 40 \text{ km/h}. \quad (43)$$

Assume the water is near 0°C , so the dynamic viscosity of water is

$$\mu = 1.787 \text{ mPa} \cdot \text{s}. \quad (44)$$

Find the thrust force required for the snowmobile to overcome the shear resistance in the thin water film and maintain its constant speed.

Solution

The shear stress in the water film is

$$\tau = \mu \frac{du}{dy}. \quad (45)$$

For a linear velocity profile,

$$\frac{du}{dy} = \frac{V}{h}. \quad (46)$$

Therefore,

$$\tau = \mu \frac{V}{h}. \quad (47)$$

Since shear stress is force divided by area,

$$\tau = \frac{F}{A}. \quad (48)$$

So,

$$\frac{F}{A} = \mu \frac{V}{h}. \quad (49)$$

Solving for F ,

$$F = A\mu \frac{V}{h}. \quad (50)$$

Now convert all quantities to SI units:

$$A = 960 \text{ cm}^2 = 960 (10^{-4} \text{ m}^2) = 0.096 \text{ m}^2, \quad (51)$$

$$h = 14 \mu\text{m} = 14 \times 10^{-6} \text{ m}, \quad (52)$$

$$V = 40 \text{ km/h} = 40 \left(\frac{1000 \text{ m}}{1 \text{ km}} \right) \left(\frac{1 \text{ h}}{3600 \text{ s}} \right) = 11.11 \text{ m/s}, \quad (53)$$

$$\mu = 1.787 \text{ mPa} \cdot \text{s} = 1.787 \times 10^{-3} \text{ Pa} \cdot \text{s}. \quad (54)$$

Substitute into the force equation:

$$F = A\mu \frac{V}{h} \quad (55)$$

$$= (0.096)(1.787 \times 10^{-3}) \left(\frac{11.11}{14 \times 10^{-6}} \right) \quad (56)$$

$$= 136.2 \text{ N}. \quad (57)$$

Final Answer

$$\boxed{F \approx 136.2 \text{ N}} \quad (58)$$

The snowmobile must generate approximately 136.2 N of thrust force to maintain a constant speed of 40 km/h over the thin water film.

16 Physical Interpretation of the Example

The snowmobile moves over a very thin water layer. The ice surface is stationary, but the sled surface moves forward with the snowmobile.

Because of the no-slip condition, the water touching the ice has zero velocity, while the water touching the sled moves with velocity V .

This velocity difference across a very small thickness creates a large velocity gradient:

$$\frac{du}{dy} = \frac{V}{h}. \quad (59)$$

A large velocity gradient creates shear stress in the water film. The snowmobile must produce thrust to overcome this shear resistance.

Important Trend

From

$$F = A\mu \frac{V}{h},$$

we can see that increasing A , μ , or V increases the required force, while increasing h decreases the required force.

17 Common Mistakes

When solving thin-film viscosity problems, students should avoid these common mistakes:

1. Forgetting to convert units such as cm^2 , μm , km/h , and $\text{mPa} \cdot \text{s}$ to SI units.
2. Using h/V instead of V/h . The velocity gradient is the change in velocity divided by the distance.
3. Confusing dynamic viscosity μ with kinematic viscosity ν .
4. Ignoring the no-slip condition at the stationary surface.

18 Summary

A fluid continuously deforms when subjected to shear stress, and the no-slip condition requires the fluid velocity at a solid surface to equal the velocity of that surface.

Key Equations

$$\begin{aligned} \rho &= \frac{m}{V}, & \gamma &= \rho g, \\ \tau &= \mu \frac{du}{dy}, & \frac{du}{dy} &= \frac{V}{h}, \\ F &= A\mu \frac{V}{h}, & \nu &= \frac{\mu}{\rho}. \end{aligned}$$